

## Do we need both views?

A prompting-strategy ablation on urban-attribute VQA

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Qwen3.5-9B-AWQ · 9 urbanization tasks · greedy decoding

# The question

- Each sample pairs a **marked satellite** image with a **4-angle street-view** grid.
- For the **9 urban-attribute tasks**, we ask: does the model actually *need both views*?
- We compare four inputs — full (both), sat only, sv only, blind (none) — under **three prompting strategies**, on the same records, greedy decoding.

Cross-view tasks (mismatch\_\*, camera\_direction) are excluded: their answer lives in *having both images*, so a single-view ablation is ill-posed there.

## Three strategies, same model

**current** thinking off, one turn, ask directly. The baseline.

**think16k** thinking *on*, up to a 16k-token <think> budget, one turn.

**multistep** thinking off, a 4-turn conversation: *observe* each image  $\rightarrow$  *reason* without options  $\rightarrow$  *self-check*  $\rightarrow$  *commit*. Turns uncapped.

$$\mathbf{synergy} = \text{full} - \max(\text{sat}, \text{sv}) \quad \mathbf{vision} = \text{full} - \text{blind}$$

## Headline: the second view earns nothing

| strategy  | full | sat  | sv   | blind | synergy | vision |
|-----------|------|------|------|-------|---------|--------|
| current   | 48.6 | 44.4 | 50.9 | 40.9  | -2.3    | +7.7   |
| think16k  | 49.8 | 46.1 | 51.8 | 41.0  | -2.0    | +8.8   |
| multistep | 51.7 | 49.4 | 53.9 | 43.4  | -2.2    | +8.3   |

- **sv-only**  $\geq$  **full** for every strategy — synergy is **negative** throughout.
- Vision helps ( $\approx +8$  over blind), but it is **one-view vision**: street-view alone.

## Per task — all strategies × all views (1/3)

| task            | strat     | full | sat  | sv   | blind | syn  |
|-----------------|-----------|------|------|------|-------|------|
| Land use        | current   | 53.0 | 50.1 | 52.7 | 43.5  | +0.3 |
|                 | think16k  | 51.8 | 45.1 | 51.5 | 51.5  | +0.3 |
|                 | multistep | 49.4 | 50.1 | 53.0 | 50.6  | -3.6 |
| Building height | current   | 61.1 | 47.9 | 64.7 | 45.8  | -3.6 |
|                 | think16k  | 62.6 | 51.6 | 61.6 | 38.4  | +1.0 |
|                 | multistep | 61.1 | 54.7 | 58.4 | 43.2  | +2.7 |
| Urban density   | current   | 38.5 | 39.2 | 44.9 | 40.4  | -6.4 |
|                 | think16k  | 38.7 | 38.0 | 43.9 | 32.3  | -5.2 |
|                 | multistep | 41.6 | 43.7 | 45.4 | 35.9  | -3.8 |

## Per task — all strategies × all views (2/3)

| task             | strat     | full | sat  | sv   | blind | syn         |
|------------------|-----------|------|------|------|-------|-------------|
| Junction type    | current   | 62.6 | 48.8 | 62.6 | 48.5  | <b>+0.0</b> |
|                  | think16k  | 57.5 | 54.8 | 59.0 | 32.3  | <b>-1.5</b> |
|                  | multistep | 63.5 | 59.3 | 62.9 | 46.4  | <b>+0.6</b> |
| Green space      | current   | 36.5 | 35.5 | 37.9 | 64.5  | <b>-1.4</b> |
|                  | think16k  | 41.4 | 38.4 | 49.8 | 59.1  | <b>-8.4</b> |
|                  | multistep | 64.5 | 62.1 | 65.0 | 68.5  | <b>-0.5</b> |
| Amenity richness | current   | 36.8 | 36.1 | 33.5 | 19.5  | <b>+0.7</b> |
|                  | think16k  | 36.3 | 37.3 | 34.9 | 20.9  | <b>-1.0</b> |
|                  | multistep | 37.1 | 38.7 | 36.1 | 25.9  | <b>-1.6</b> |

## Per task — all strategies × all views (3/3)

| task            | strat     | full | sat  | sv   | blind | syn          |
|-----------------|-----------|------|------|------|-------|--------------|
| Road type       | current   | 47.5 | 40.1 | 57.5 | 34.9  | <b>-10.0</b> |
|                 | think16k  | 58.4 | 50.4 | 64.4 | 34.4  | <b>-6.0</b>  |
|                 | multistep | 53.4 | 48.5 | 63.2 | 33.3  | <b>-9.8</b>  |
| Road surface    | current   | 82.2 | 78.9 | 87.1 | 67.7  | <b>-4.9</b>  |
|                 | think16k  | 83.2 | 75.2 | 84.8 | 94.4  | <b>-1.6</b>  |
|                 | multistep | 89.8 | 80.9 | 89.4 | 83.8  | <b>+0.4</b>  |
| Transit density | current   | 32.1 | 30.9 | 30.4 | 27.3  | <b>+1.2</b>  |
|                 | think16k  | 31.6 | 33.3 | 31.6 | 26.6  | <b>-1.7</b>  |
|                 | multistep | 29.7 | 26.8 | 31.4 | 27.8  | <b>-1.7</b>  |

## Takeaway

- On all **9 urbanization tasks**, across all **3 strategies**, full never beats the best single view by a meaningful margin.
- The **satellite view is redundant** given the street-view: sv-only matches or beats the two-view setup everywhere.
- Better prompting raises the ceiling (multistep +3.1 over baseline full), but it lifts sv-only *in lockstep* — the gap never closes.
- **Conclusion:** for these tasks we do *not* need both views. One well-chosen view (street-level) carries the signal; the second image does not earn its cost.